

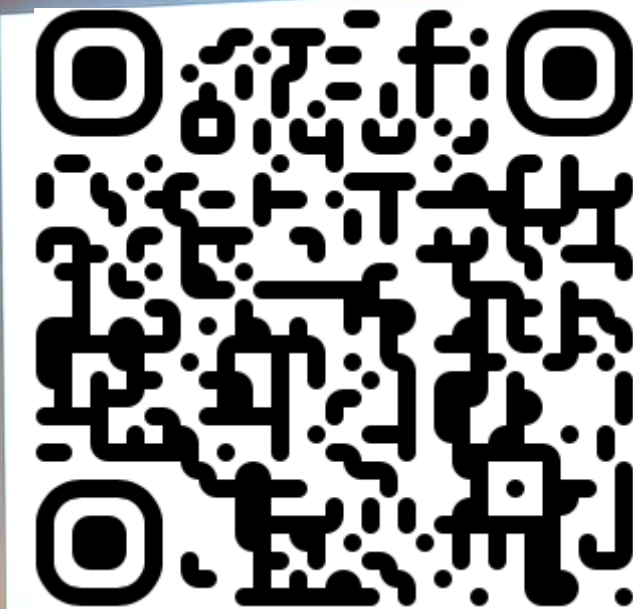


Software Security Issues for Small IoT SoCs

Stephen Cravey, MSc, CISSP



International
Ecology
Symposium
2024



About

- Apple II
- Software
- Electronics
- 2600
- Unix
- ECE/VLSI
- ISP Architect
- BGP Monkey
- DFIR
- Military/Defense
- NASA ISS Program
- Recovering CISO
- Big Consulting
- cravey@gmail.com

Foundations – Internet

- Network of Networks



Foundations – IoT



- Coined ~2005 by ITU
- Things are EVERYWHERE!

Foundations - IoT Examples



Foundations - IoT Examples

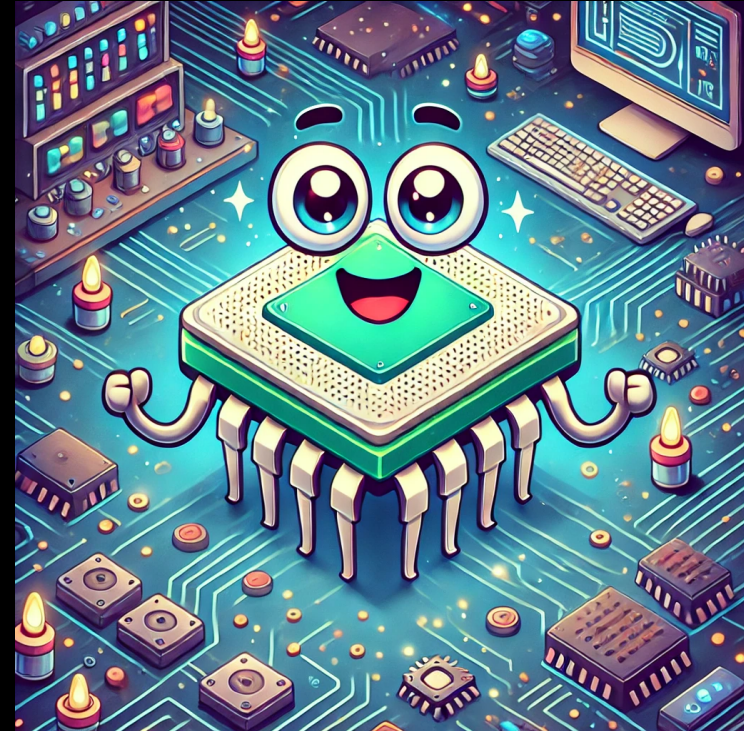


Foundations - IoT Examples



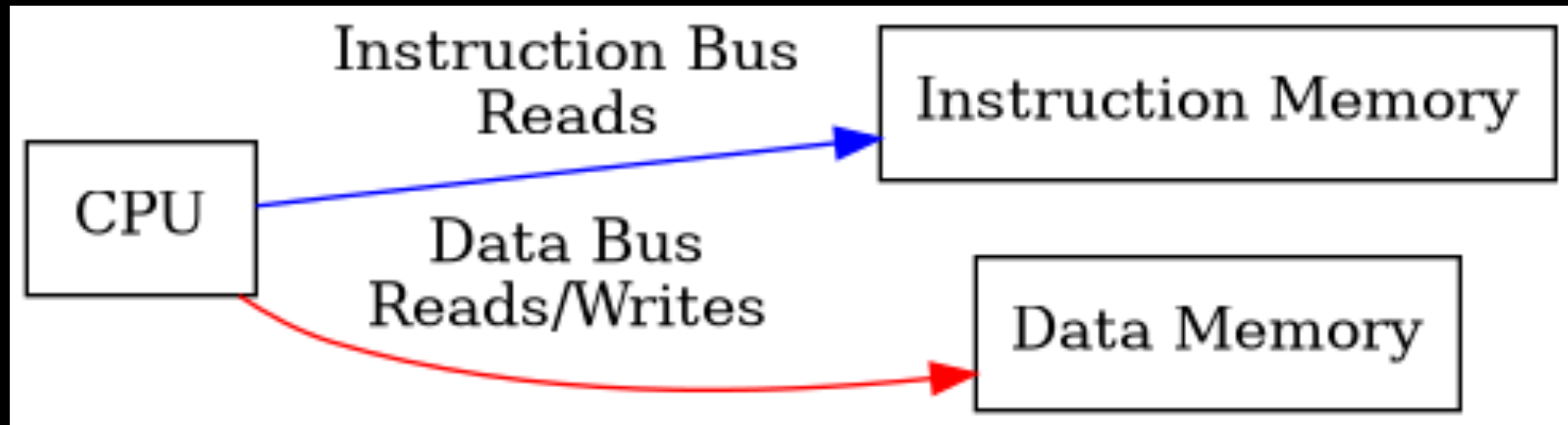
Foundations – Microprocessors

- Takes INPUT
 - Performs some operation (Processes INPUT)
 - Provides OUTPUT
-
- <https://itemas.github.io/6502js/>

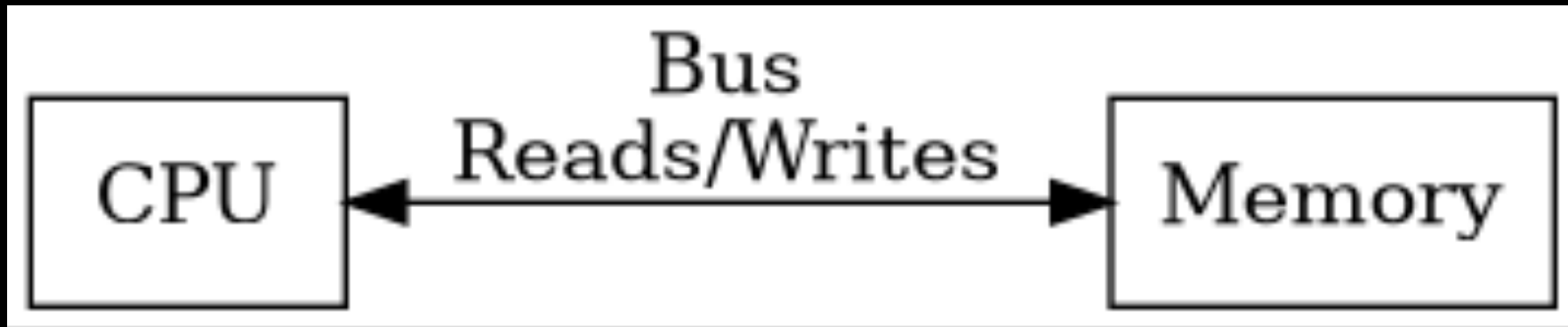


Foundations – Microprocessors

Harvard Architecture



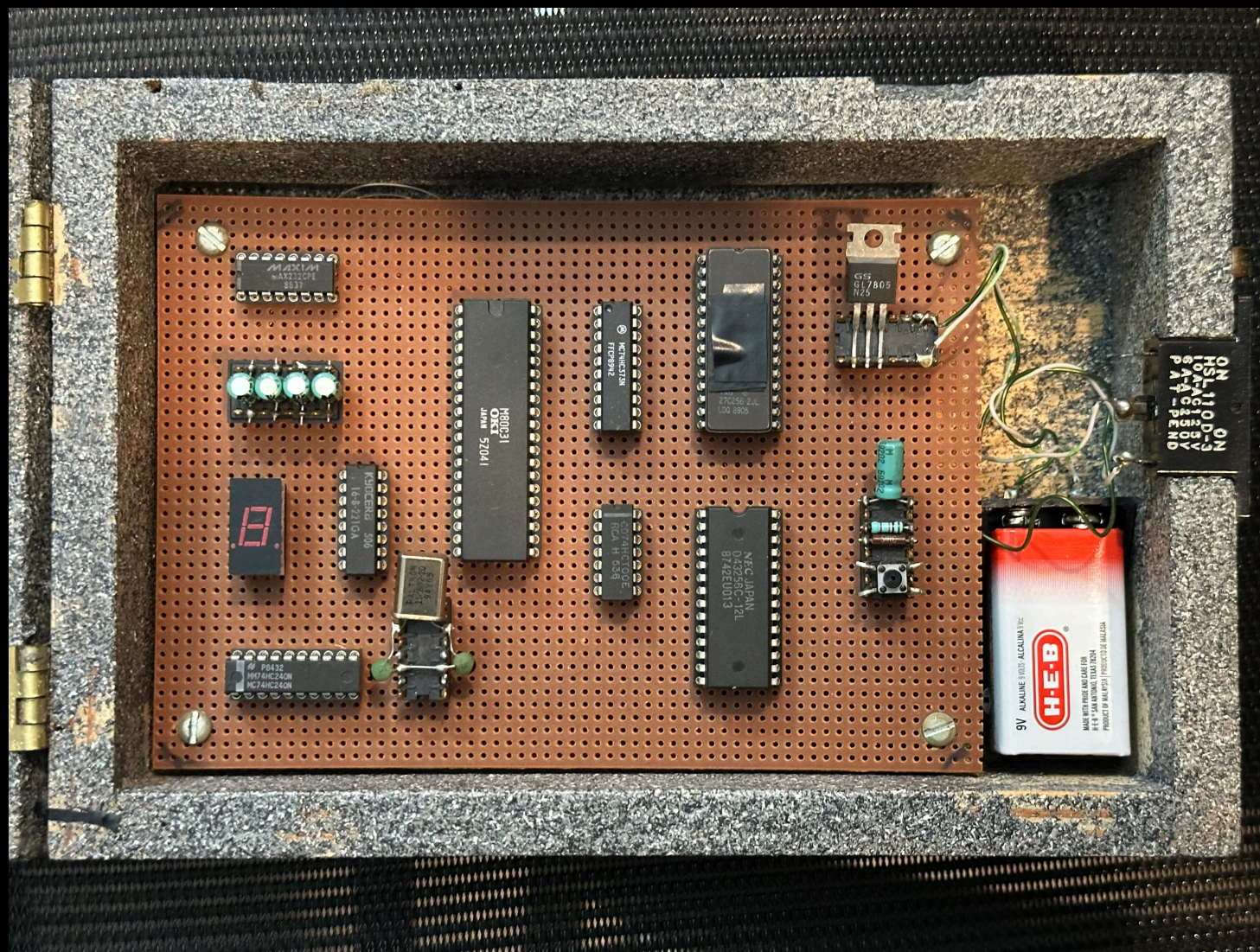
Foundations – Microprocessors von Neumann Architecture



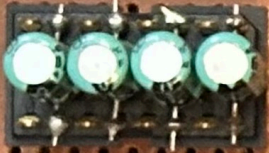
Foundations – SoCs



Foundations - SoCs



MAXIM
MAX232CPE
8637



P8432
MM74HC240N
MC74HC240N

K90CENR 306
16-B-221GA



M80C31
OKI
JAPAN 5Z041

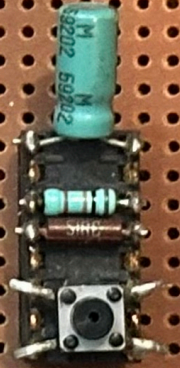
MC74HC373N
FTCP8942

CD74HC100E
RCA H 638

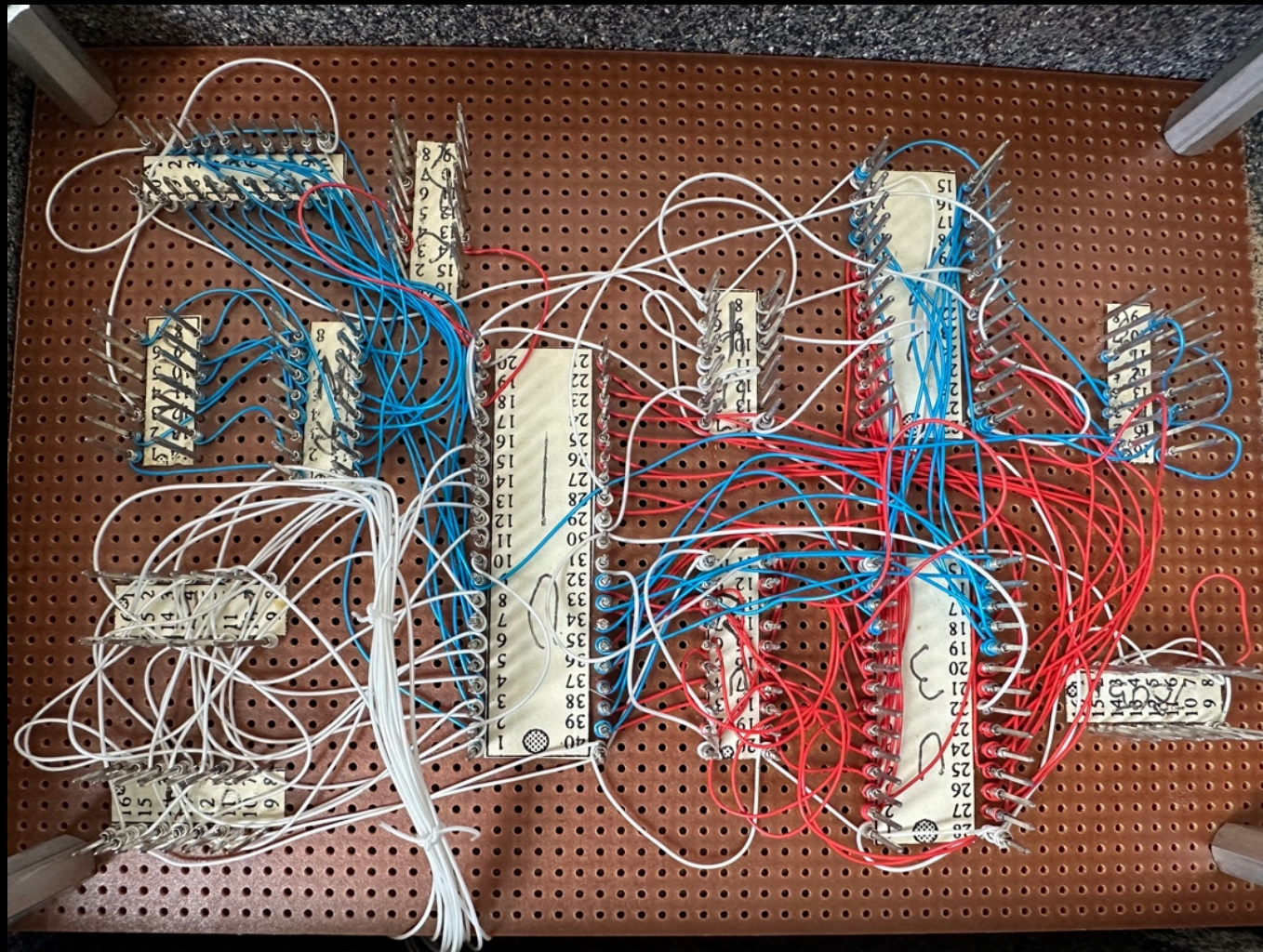
27C256-2JL
LDQ 8905

NEC JAPAN
D43256C-12L
8742EU013

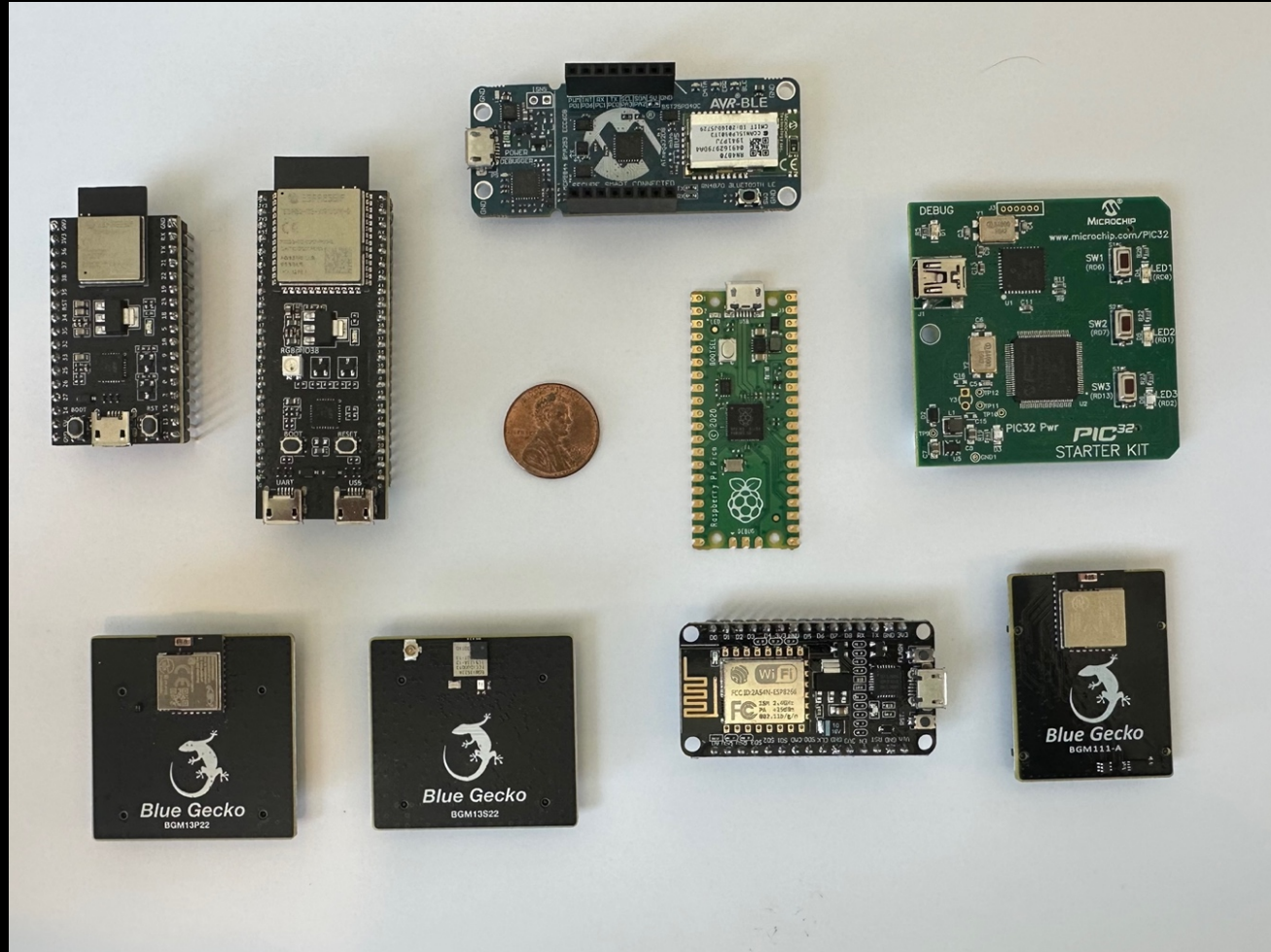
GS
GL7805
N25



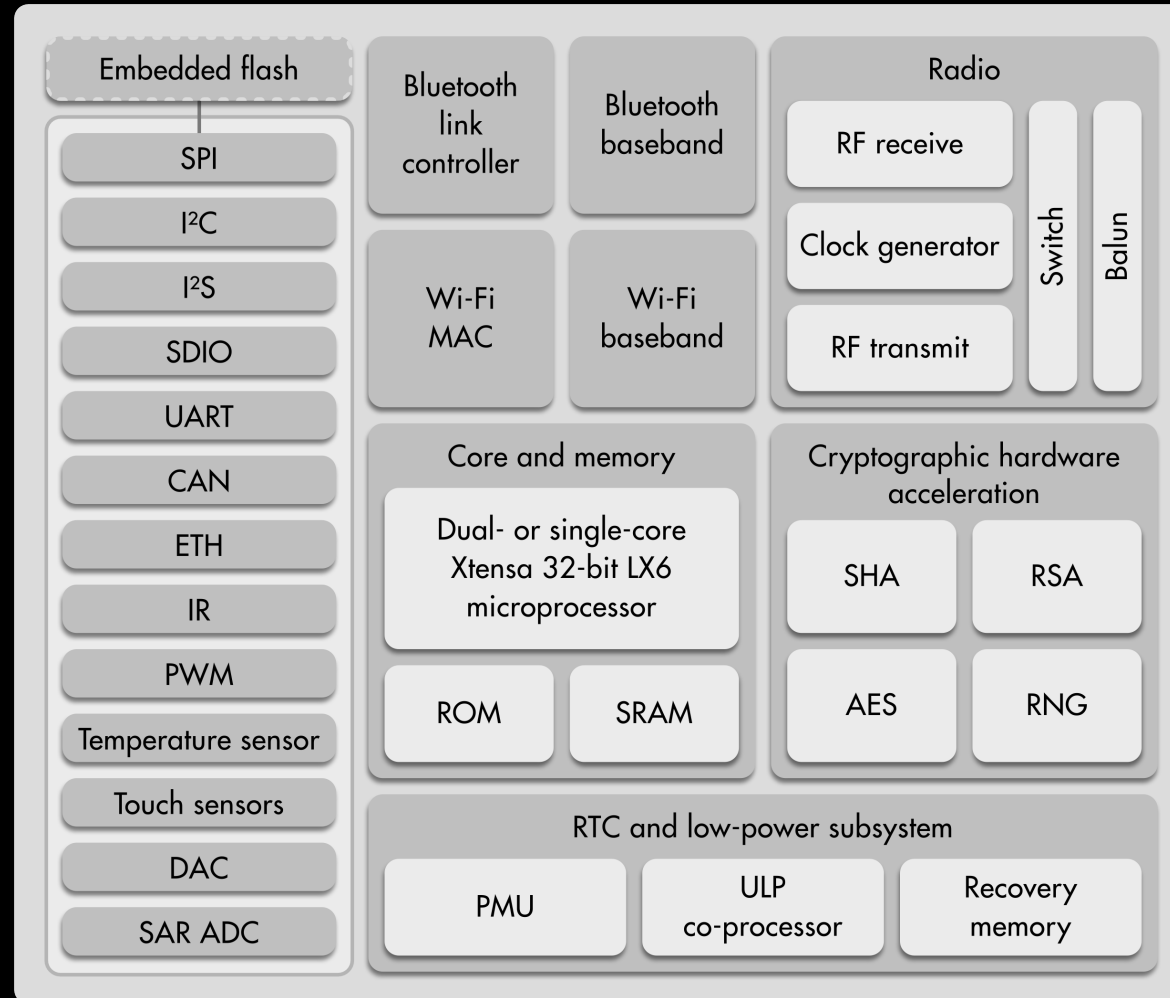
Foundations – SoCs



Foundations – SoC Drawer



Foundations – SoC Block Diagram



Foundations – Binding Software to Hardware

- Software can only run on approved hardware
- Copy Protection, etc.

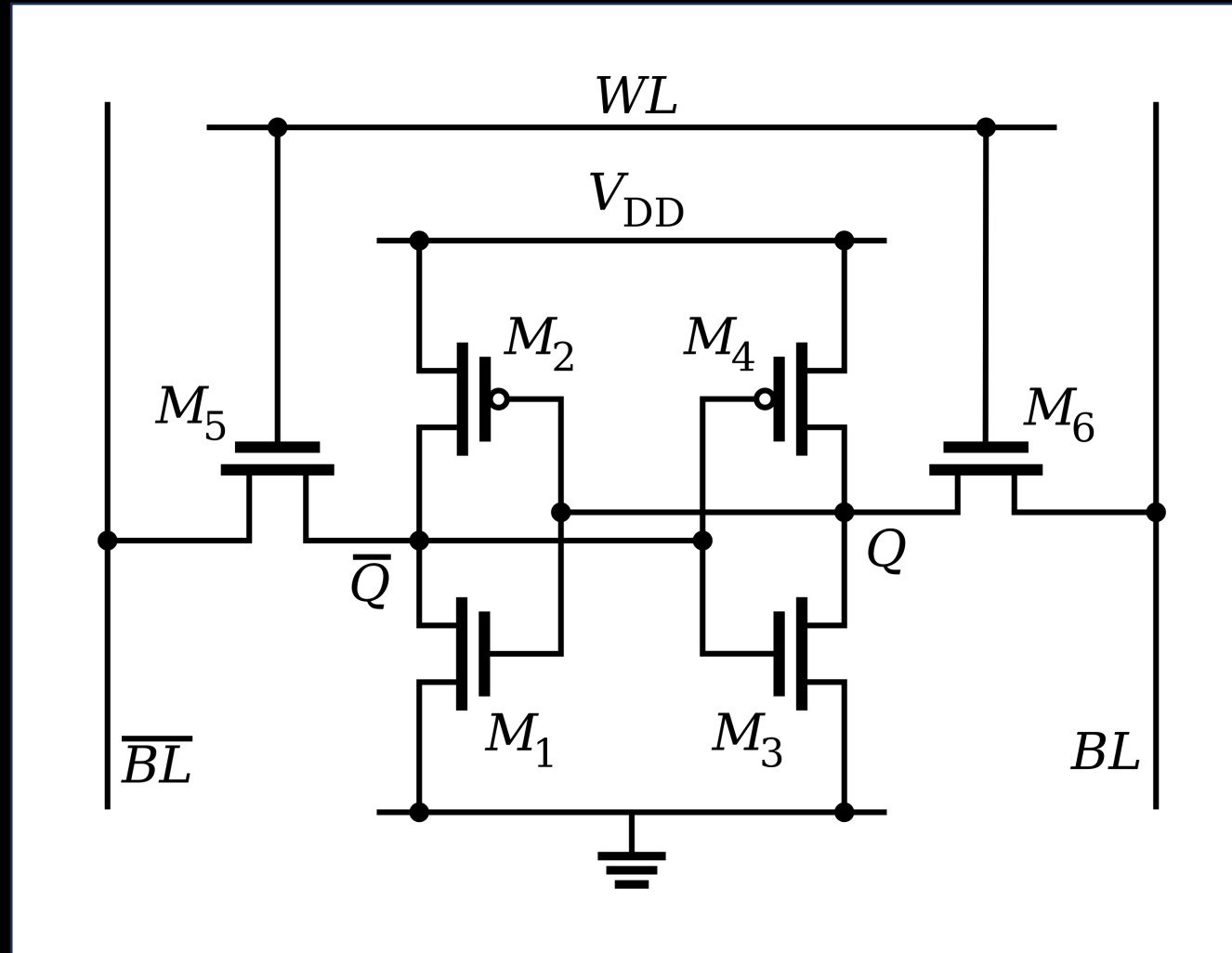
Foundations – Binding Hardware to Software

- Prevents Hardware from executing unapproved software

Foundations – PUFs

- Physically Unclonable Functions?
- Physically Unclonable Features?
- Physical Unclonable Features?
- Referred to as a “*-PUF”

Foundations – SRAM PUF



Foundations – HSM

- Hardware Security Modules
 - Provide basic cryptographic capabilities.
 - Unique Identifiers
 - TRNG
 - Misc Other Features

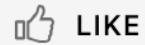
SoC Selection Criteria

- Harvard Architecture
- Distinct ISAs
- **No NDA for technical data**
- Something I've encountered in my daily life

SoCs Selected

ESP32 IoT Devices Vulnerable to Forever-Hack

DEC 02, 2019 • 2 MIN READ



by



Erik Costlow

[FOLLOW](#)

Product Mgmt @ Azul. Cryptography
course author.

A security researcher has identified a local-access technique that enables [a complete bypass of all security within the popular ESP32 IoT chip](#), where malware can be implanted and never removed. Espressif has [confirmed the vulnerability with an advisory](#). The same attack also extracts cryptographic keys and works against the device's most secure configuration.

SoCs Selected

It looks like the Raspberry Pi RP2350 Hacking Challenge may have been beaten — Hacker gains access to the OTP secret by glitching the RISC-V cores to enable debugging

News

By [Mark Tyson](#) | Published January 4, 2025

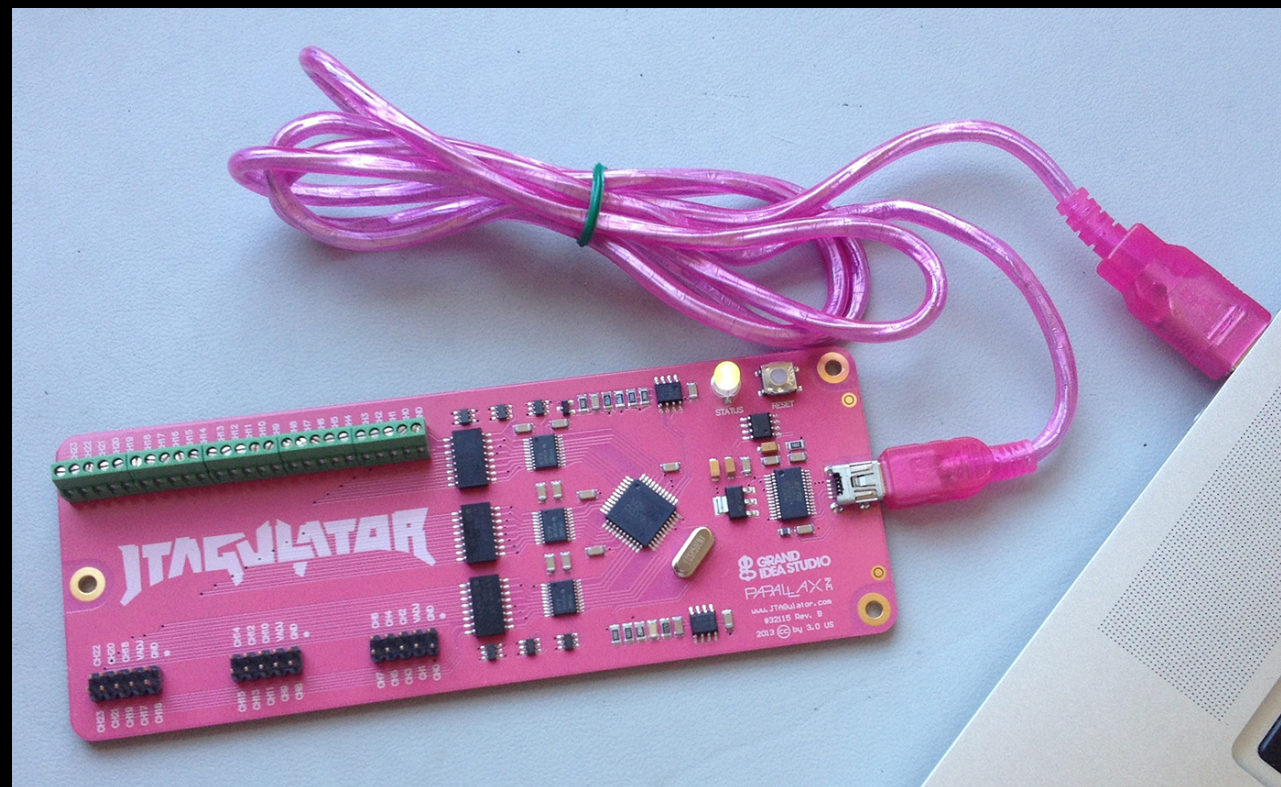
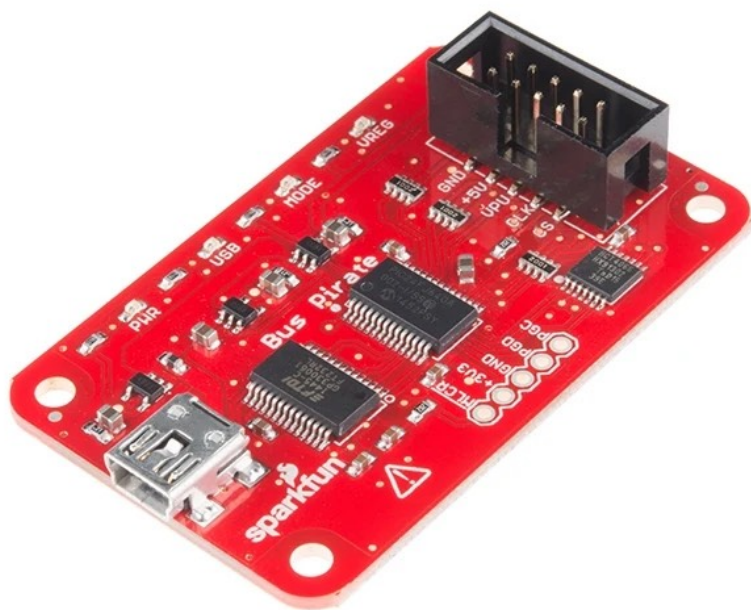
Engineer Aedan Cullen may be in line for the \$20K prize.

Lessons Learned

- For small, common IoT SoCs, security is often not a priority.
- More expensive SoCs are often better.
- Devtools often kindof suck.
- Quality embedded development is harder than expected.
- Open season on many IoT devices!



Tools

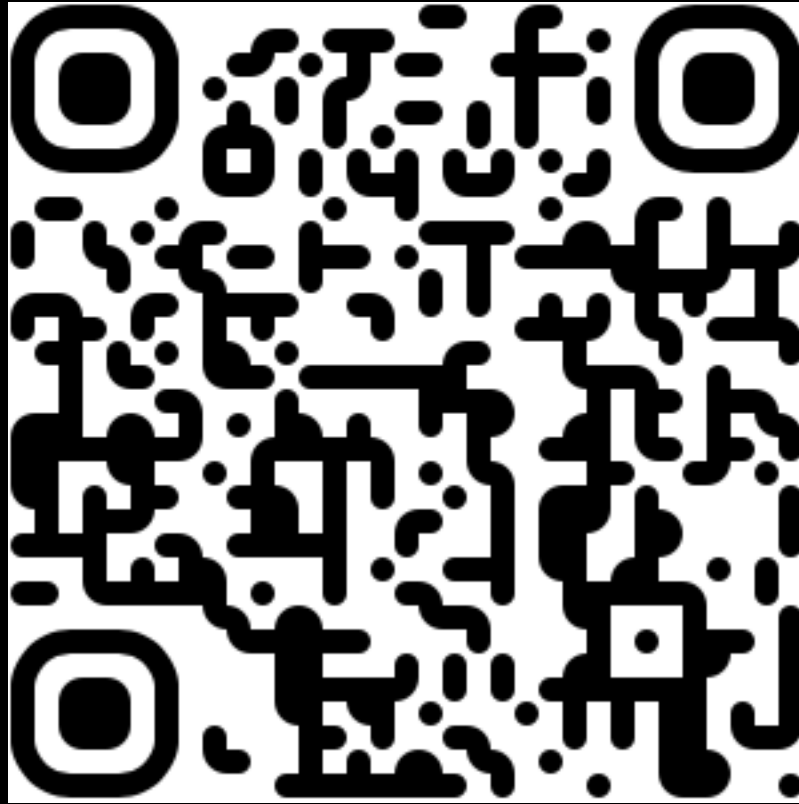


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- Mostly overwhelmed with Spam



SETEC Astronomy



A Brief Introduction to Cognitive Warfare

